



Introduction to Epidemiology

APHS 20203 • Epidemiology & Biostatistics

Course Learning objectives

1. Define epidemiology and key associated terminology, especially confounding.
2. Distinguish among association, prediction, and causation.
3. Explain the Rothman model of sufficient-component causes (RMSCC) and the concept of multifactorial causation.
4. Explain the limitations of anecdotal evidence in drawing epidemiologic conclusions.
5. Explain why disseminating risk information is sometimes necessary but rarely sufficient to improve population health.

Lecture Learning objectives

By the end of this lesson, you'll be able to

1. Define epidemiology and explain its core assumptions.
2. Describe key events in the history and development of epidemiology.
3. Differentiate epidemic, endemic, and pandemic patterns of disease occurrence.
4. Describe at least three applications of epidemiology in public health practice.

What is Epidemiology?

Respond here

What is Epidemiology?

EPIDEMIOLOGY

- epi – “on, upon”
- demo – “people”
- ology – “study of”

Definition: the study of the distribution and determinants of health-related states or events (e.g. infectious disease) in specified (human) populations and the application of this study to control health problems

-- James Last, 1995, *A Dictionary of Epidemiology*

The Epidemiologic Triangle

Scenario Exercise

1. **Scenario:** A city health department notices that one neighborhood has twice the rate of asthma ER visits compared to another neighborhood in the same city. This difference has been observed every year for the past three years.
2. **Questions:** Do you think this difference is mostly due to random chance—yes or no? If it's not random, what is one thing that might explain the difference?
3. **What are we assuming if we believe those factors matter?**

Assumptions of Epidemiology

ASSUMPTION 1

Human disease does not occur at random

Disease frequency and distribution vary because of identifiable factors that increase or decrease risk.

ASSUMPTION 2

Determinants of disease can be identified

Causal and preventive factors can be studied through systematic investigation of populations.

Descriptive and Analytic Epidemiology

Descriptive Epidemiology Is A Necessary Antecedent Of Analytic Epidemiology

DISTRIBUTION

Descriptive epidemiology

- When was the population affected?
 - Where was the population affected?
 - Who was affected?
-

DETERMINANTS

Analytic epidemiology

- How was the population affected?
- Why was the population affected?

Cholera Outbreak In London

Cholera Symptoms

Cholera is an extremely virulent disease that can cause severe acute watery diarrhea. Cholera affects both children and adults and can kill within hours if untreated.

It takes between 12 hours and 5 days for a person to show symptoms after ingesting contaminated food or water.



Book cover for The Lambeth Cholera Outbreak of 1848–1849 by Amanda J. Thomas.

Micro-Activity: Thinking Like an Epidemiologist

1. Watch the short video on John Snow and cholera. **John Snow: Pioneer of Epidemiology**
2. As you watch, focus on these questions:
What patterns did Snow observe? (*Who, where, when — descriptive epidemiology*)
3. What explanation was he testing? (*How, why — analytic epidemiology*)

Descriptive epidemiology comes first, analytic epidemiology builds on it.

History of Epidemiology: Key Events

Event	Time Period	Description
Bubonic Plague Epidemics (Black Death)	1347 - 1351	One of the deadliest pandemics, causing 75–200 million deaths in Eurasia.
Development of Toxicology	16th Century	Paracelsus, "Father of Toxicology," introduced the principle: "The dose makes the poison."
Development of Biostatistics	18th Century	John Graunt pioneered statistical analysis, using Bills of Mortality to study disease patterns in London.
Development of Smallpox Vaccine	1796	Edward Jenner created the first

Event	Time Period	Description
		vaccine by using cowpox to confer immunity to smallpox.
1918 Influenza Pandemic	1918-19	The "Spanish flu" infected a third of the global population, killing 50–100 million.
Identification of Smoking as a Cause of Cancer	1950	Landmark studies linked smoking to lung cancer, driving public health campaigns against tobacco use.
Eradication of Smallpox	1980	WHO declared smallpox eradicated, marking the success of global vaccination efforts.

Renowned Epidemiology Scholars

How Epidemiologists Learned to Think:

Hippocrates → Environment and behavior influence health

John Graunt → Counting populations reveals disease patterns

William Farr → Vital statistics and surveillance systems

John Snow → Using population data to identify causes

Alexander Louis → Quantitative reasoning in medicine

Bradford Hill → Systematic thinking about causality

The Contemporary Era: 1940 to Present

01

Long-term population studies

Framingham Heart Study → identification of risk factors

02

Expanded causal thinking

Chronic disease linked to behaviors, infections, and genetics

03

Ethics and equity in research

William Carter Jenkins → accountability in public health studies

From History to Practice

Epidemiology began by asking:

Who is affected?

Where does disease occur?

When does it change over time?

Next: Applying these questions to real data
(Person, Place, and Time)

Who: personal characteristics e.g. age or gender

 Line chart of breast cancer death rates by age group for all persons in the United States in 1996 and 2006.

 Bar chart showing the percent of female breast cancer deaths by age group.

Age-Specific Death Rates Associated With Breast Cancer Among All Persons by Age Group, United States, 1996 and 2006

Note: The death rates refer to the number of deaths due to breast cancer per 100,000 persons in the particular age group (including both females and males). Source: National Center for Health Statistics. Available at <http://www.prb.org/Articles/2009/breastcancer.aspx?p=1>

Where: Geographically restricted or widespread (pandemic)?

*Rates are per 100,000 and are age-adjusted to the 2000 U.S. standard population. †Data are suppressed at the state's request. †Source: U.S. Cancer Statistics Working Group. United States Cancer Statistics: 1999–2009 Incidence and Mortality Web-based Report. Atlanta (GA): Department of Health and Human Services, Centers for Disease Control and Prevention, and National Cancer Institute; 2013. Available at: <http://www.cdc.gov/cancer/breast/statistics/state.htm>

Female Breast Cancer Incidence Rates* by State, 2009†

When: change over time

- Rates per 100,000 women aged 45–64 years for whom breast cancer was the underlying cause of death (based on International Classification of Diseases, Ninth Revision [ICD-9] codes 174–175 for 1990–1998 and ICD-10 code C50 for 1999–2007). Sources: CDC Morbidity and Mortality Weekly Report (MMWR), July 30, 2010 / 59(29);915. Available at <http://www.cdc.gov/mmwr/preview/mmwrhtml/mm5929a5.htm>

Breast Cancer Death Rates Among Women Aged 45–64 Years,* by Race — United States, 1990–2007

From Patterns to Classification

01

Endemic

02

Epidemic

03

Pandemic

We use person, place, and time to describe disease patterns. These patterns are commonly classified as endemic, epidemic, and pandemic. These terms describe patterns over time and place, not disease severity.

Disease Distribution Patterns

01

Endemic

Disease or condition consistently present in a population

- Relatively stable over time
 - Expected level of occurrence in a given place
-

02

Epidemic (Outbreak)

Disease occurrence in excess of what is normally expected

- Sudden increase over a short period
 - Often limited to a specific region
-

03

Pandemic

An epidemic that spreads across multiple countries or continents

- Global distribution
- Affects a large proportion of the population

Epidemic: Visual Illustration

Global Context: Epidemics and Pandemics

Exercise: Apply the Concepts

Match each term with the correct example.

A. endemic B. pandemic C. epidemic

1. Malaria is always present in Africa due to the presence of infected mosquitoes. Malaria is _____ in Africa.
2. Ebola virus cases in parts of Africa exceed what is normally expected for that region. This virus represents an_____.
3. COVID-19 spread across countries and continents worldwide. COVID-19 is a_____.

Exercise: Apply the Concepts

Match each term with the correct example.

A. endemic B. pandemic C. epidemic

1. Malaria is always present in Africa due to the presence of infected mosquitoes. Malaria is **endemic** in Africa.
2. Ebola virus cases in parts of Africa exceed what is normally expected for that region. This virus represents an **epidemic**.
3. COVID-19 spread across countries and continents worldwide. COVID-19 is a **pandemic**.

Exercise: Check Your Understanding

TRUE OR FALSE

1. A pandemic is always a form of epidemic.
2. An endemic disease always represents an epidemic.

Pause & think

Exercise: Check Your Understanding

1. A pandemic is always a form of epidemic.

- Answer: True
- Explanation: A pandemic is an epidemic that has spread across multiple countries or continents. It still meets the definition of an epidemic—occurrence in excess of what is normally expected—but on a global scale.

2. An endemic disease always represents an epidemic.

- Answer: False
- Explanation: An endemic disease is consistently present at an expected level within a population or region. Because it does not exceed normal expectations, it is not considered an epidemic, even though cases may occur continuously.

Use of Epidemiology

In this course Epi Group Presentation, you will:

Describe a health problem

(Who is affected? Where? When?)

Explain possible reasons

(Risk factors or exposures)

Interpret evidence

(Patterns and associations)

Propose prevention or control strategies

Revisit Learning objectives

1. Unpack the definition and assumptions of epidemiology

1. Identify epidemiology key events
2. Distinguish between disease distribution patterns: epidemics, endemics, and pandemics
3. Describe at least three uses of epidemiology



